



Alberta Public Health Disease Management Guidelines

Zika Virus



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Alberta Health

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Case Definition

Confirmed Case

Laboratory confirmation^(A) of infection with or without clinical illness^(B) **AND** at least one epidemiological linkage^(C) identified:

- Detection of Zika virus (ZIKV) nucleic acid (e.g., by RT-PCR) or isolation from an appropriate clinical specimen (e.g., blood, CSF, urine);

OR

- Positive ZIKV IgM antibody with or without ZIKV IgG seroconversion **AND** a positive ($\geq 1:20$) ZIKV Plaque Reduction Neutralization Test (PRNT) where the DENV PRNT is negative (refer to Table 1 for more information);^(D)

Probable Case

Laboratory evidence^(A) of infection **AND** at least one epidemiological linkage^(C) identified:

- Positive ZIKV IgM antibody with or without ZIKV IgG and PRNT for ZIKV or DENV was not tested (refer to Table 1 for more information).^(D)

^(A) Due to the cross-reactivity of flaviviruses, Zika serology needs to be interpreted in conjunction with dengue serology, when available.

^(B) Clinical illness: refer to [Clinical Presentation](#).

^(C) Epidemiological linkage:⁽²⁴⁾

- Resides in or recent travel to an area with known ZIKV transmission (See [CDC Yellow Book webpage on ZIKV](#)), OR
- Sexual contact with a confirmed or probable case within the infection transmission risk window of ZIKV infection or person with recent travel to an area with known ZIKV transmission, OR
- Vertical transmission where the mother has history of travel or residence in an area where Zika virus vectors were present during pregnancy, OR
- Receipt of blood or blood products within 30 days of symptom onset, OR
- Organ or tissue transplant recipient within 30 days of symptom onset, OR
- Association in time and place with a confirmed or probable case, OR
- Likely vector exposure in an area with suitable seasonal and ecological conditions for potential local vector borne transmission.

^(D) Due to declining ZIKV activity worldwide, PRNT testing for flaviviruses such as dengue and Zika will only be requested when ZIKV is suspected (usually in congenital cases). Refer to Table 1 for assistance with interpreting lab results and/or consult the Virologist-on-call.

Table 1: Zika and Dengue Lab Result Interpretation⁽¹⁾*(Highlighted light blue are the most important when reviewing results)*

NOTIFIABLE							
NAT/PCR Testing	Zika Testing			Dengue Testing			Interpretation
	ZIKV IgM	ZIKV IgG	ZIKV PRNT	DENV IgM	DENV IgG	DENV PRNT	
ZIKV +	Any result ^(C)	Any result	Any result	Any result	Any result	Any result	CONFIRMED ZIKA
Negative Not tested	Positive	Any result	≥20 (Positive)	Any result	Any result	Negative	CONFIRMED ZIKA (dengue – not a case)
Negative Not tested	Positive	Any result	Not tested	Negative, Not tested	Negative, Not tested	Not tested	PROBABLE ZIKA
DENV +	Any result	Any result	Any result	Any result	Any result	Any result	CONFIRMED DENGUE
Negative Not tested	Not tested	Not tested	Not tested	Positive	Sero-conversion	Not tested	CONFIRMED DENGUE
Negative Not tested	Any result	Negative, Not tested	Negative^(D)	Positive	Any result	≥20 (Positive)	CONFIRMED DENGUE (zika – not a case)
Negative Not tested	Negative, Not tested	Negative, Not tested	Not tested	Positive	Any result	Not tested	PROBABLE DENGUE However, if subsequent testing show the following results, this indicates NOT A CASE : e.g., DENV IgM positive/IgG negative (no seroconversion) OR DENV IgM negative /IgG negative (false positive IgM).
Negative Not tested	Any result	Any result	≥20 (Positive)	Any result	Any result	≥20 (Positive)	PROBABLE FLAVIVIRUS – specific virus cannot be identified due to cross-reacting flavivirus antibodies. Refer to Flavivirus case definition.

NOT NOTIFIABLE – For information only to assist with case management and classification						
ZIKV IgM	ZIKV IgG	ZIKV	DENV IgM	DENV	DENV PRNT ^(F)	Interpretation
Pos, Neg ^(F) or Equiv ^(E)	Pos or Neg	Pending	Pos, Neg or Equiv	Any result	Pending	SUSPECT CASE: zika or dengue, pending lab results
Any result	Pos or Neg	Neg	Pos, Neg or Equiv	Any result	Neg	NOT A CASE: No evidence of zika or dengue, however if a specimen was collected <5 days after onset the PRNT results may be negative. Convalescent specimens are recommended in 14 days.
Equiv, Neg or Not tested	Pos or Neg	≥20 (Positive)	Neg	Any result	Neg	NOT A CASE: Evidence of zika infection; likely remote infection > 3months (Not an acute case)
Equiv, Neg or Not tested	Pos or Neg	Negative	Equiv, Neg or Not tested	Any result	≥20 (Positive)	NOT A CASE: Evidence of dengue infection; likely remote infection > 3months (Not an acute case)
Not tested			Neg or Equiv ^(E)	Any result	Not tested	NOT A CASE: Equivocal, indeterminate or inconclusive results indicate that result could not be interpreted as positive or negative. New samples should be collected if clinically indicated

^(C) "Any result" includes "Not tested". Applicable where ever this phrase is used in the table.^(D) For the purposes of lab result interpretation, a negative PRNT is any value less than 20.^(E) Equivocal, indeterminate and inconclusive results are indicative of laboratory samples that could not be interpreted for various reasons (e.g., borderline positive or non-specific binding) in order to give a result of positive or negative. If clinically indicated, new specimens should be collected for these individuals. Applicable where ever this phrase is used in the table.^(F) Individuals tested >3 months after exposure will almost always test IgM negative. If clinically indicated (e.g., pregnancy) samples may be forwarded to NML for PRNT testing with incomplete serology.

Reporting Requirements

Physicians, Health Practitioners and Others

- Physicians, health practitioners or others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

- Laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the MOH (or designate) of the zone and the Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations Inuit Health Branch Public Health

- The MOH/designate of the zone where the case currently resides shall report all confirmed and probable cases to the CMOH/designate using the Notifiable Disease Report (NDR) form or another mutually agreed upon reporting system within two weeks of notification for the initial NDR and the final NDR (amendment) within four weeks, as per usual.
- For out-of-province and out-of-country cases, the following information should be forwarded to the CMOH/or designate as per usual communicable disease reporting processes and include:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - any positive laboratory report/s, and
 - other relevant clinical / epidemiological information.

Epidemiology

Etiology

Zika virus (ZIKV) is a member of the *Flaviviridae* family.⁽²⁾

Clinical Presentation

The majority of individuals infected with ZIKV will have no symptoms; however, approximately 20–25% will develop symptoms.⁽²⁾ Symptoms are usually mild and can include fever, rash, conjunctivitis, myalgia, arthralgia and headache lasting from 2 – 7 days.⁽³⁾ Most people recover without incident or hospitalization.

Complications of ZIKV include Guillain-Barré syndrome or other neurological manifestations such as acute myelitis, meningoencephalitis and acute disseminated encephalomyelitis.^(4–6) ZIKV has been linked to microcephaly and other congenital anomalies in infants infected with ZIKV during pregnancy.⁽⁷⁾

Diagnosis

PCR testing by ProVLab should be performed on plasma (blood in EDTA) collected during the first two weeks after symptom onset.⁽⁸⁾ ZIKV has been detected by PCR in urine for 2 – 3 weeks and in semen up to 6 months.

Antibody detection should be performed on whole blood in SST tubes.⁽⁹⁾ ZIKV IgM antibodies are usually detectable 6 – 7 days after the onset of illness and generally indicate a current or recent infection. A serum specimen should be collected for acute phase ZIKV IgM antibody testing at least 5 days and up to 12 weeks after the onset of clinical illness. A convalescent phase serum should be obtained at least 2 weeks after the first sample has been collected. As is the case for DENV virus, ZIKV IgM antibodies will likely not be detected beyond 3 months after exposure.

Currently, the most frequently acquired flavivirus in returning travelers is dengue. Flavivirus antibodies can cross-react in the existing serological tests, especially those for IgM, resulting in some uncertainty in IgM results between ZIKV and DENV. If ZIKV is suspected, cases with positive or equivocal serology are sent to the National Microbiology Laboratory (NML) in Winnipeg to be confirmed using PRNT. This is mainly indicated in scenarios where congenital ZIKV infection is suspected after consultation with the Virologist-on-call.

PRNT is not always able to provide a definitive determination of the specific flavivirus causing a recent infection, especially if the person has a prior history of flavivirus infection (e.g., previous DENV or West Nile virus), is immunized against a flavivirus (e.g., yellow fever or Japanese encephalitis) or is currently co-infected with another flavivirus. These cases may have test results such as DENV IgM negative with positive PRNT results or positive PRNT results for both DENV and ZIKV, making it difficult to determine which virus is causing the illness (i.e., probable flavivirus).

(Refer to the [ProvLab website](#) for the latest information regarding diagnosis and testing).

Treatment

- There is no effective therapy for ZIKV. Supportive care only.
- ZIKV infections have not shown the same bleeding risks as DENV but should be managed the same way until DENV is ruled out. Acetylsalicylic acid (ASA) and other non-steroidal anti-inflammatory drugs (NSAIDs) should not be used, if possible, as they may aggravate the bleeding.⁽¹⁰⁾

Reservoir

Non-human and human primates are likely the main reservoirs of the virus with anthroponotic (human-to-vector-to-human) transmission occurring during outbreaks.⁽¹¹⁾ ZIKV virus is transmitted to humans primarily through the bite of an infected *Aedes* species mosquito (*Ae. aegypti* and *Ae. albopictus*). There is no significant evidence that the *Culex* species are able to transmit the virus.

Populations of *Aedes (aegypti or albopictus)* mosquitoes are not established in Canada at this time. The likelihood that *Aedes* mosquitos will become established in Canada is considered very low as they are limited by the lack of environmental and climate suitability here.⁽²⁾

Transmission

ZIKV is most commonly transmitted to humans by the bite of an infected mosquito.⁽¹²⁾ Other less common modes of transmission include:^(13,14)

- sexual,
- vertical,
- blood and organ donation, and
- laboratory-acquired.

In addition, non-sexual person-to-person transmission has been reported in the setting of an immunocompromised contact.⁽¹⁵⁾

Similar to other flaviviruses, ZIKV is susceptible to common disinfectants and various methods of inactivation, such as ultra-violet light and heat.⁽¹⁶⁾

Incubation Period

The incubation period ranges from 3 – 12 days.⁽³⁾

Period of Communicability

ZIKV is thought to be communicable, via blood (viremia) from 3 – 7 days after the onset of symptoms.⁽²⁾ However, ZIKV has been detected in urine 2 – 3 weeks after the onset of symptoms⁽¹⁷⁾, in vaginal fluid 11 days after the onset of symptoms⁽¹⁸⁾ and in semen up to 6 months after the onset of symptoms.^(19–21)

In Canada, there is a very low risk of ZIKV transmission to mosquitos from humans during the viremic period mainly because the mosquitos that are capable of reproducing the virus are not endemic to Canada. For more information see *Reservoir*.

Host Susceptibility

Susceptibility is universal. Individuals with co-morbidities may be at increased risk for more serious ZIKV-associated complications (see Clinical Presentation).⁽²⁾ Fetuses are susceptible in utero via vertical transmission from an infected mother.⁽¹⁷⁾

Incidence

ZIKV is not endemic in Alberta and infections are likely to be acquired through international travel. ZIKA was first reported in Alberta in 2013 in a female who had travelled to Thailand.⁽²²⁾ Since December 2015, Alberta has seen additional cases related to the increase in global ZIKV disease incidence and international travel of Albertans. For more information on the occurrence of ZIKV worldwide refer to the [CDC Yellow Book webpage on ZIKV](#).

Public Health Management

Key Investigation

- Confirm the diagnosis, if possible. Other arboviral diseases such as chikungunya, DENV, West Nile virus and rash illnesses such as measles, rubella, parvovirus, enterovirus, or febrile illnesses such as malaria should be ruled out.⁽²³⁾
- Determine where the case may have been exposed, taking into consideration the incubation period, reservoir, and mode of transmission, including a history of:⁽²⁴⁾
 - recent travel to or residence in an area with known ZIKV transmission (See [CDC Yellow Book webpage on ZIKV](#)), OR
 - Sexual contact with a confirmed or probable case within the infection transmission risk window of ZIKV infection or person with recent travel to an area with known ZIKV transmission, OR
 - Vertical transmission where the mother has history of travel or residence in an area where Zika virus vectors were present during pregnancy, OR
 - Receipt of blood or blood products within 30 days of symptom onset, OR
 - Organ or tissue transplant recipient within 30 days of symptom onset, OR
 - Association in time and place with a confirmed or probable case, OR

- Likely vector exposure in an area with suitable seasonal and ecological conditions for potential local vector borne transmission.

Management of a Case

- The following information should be provided to cases as appropriate:
 - Provide education on how to prevent [transmission](#) and infection (i.e., preventive measures below).
 - Contact health care provider if symptoms worsen (refer to [Clinical Presentation](#)).
 - In general, ZIKV cases should use condoms during sex to prevent disease transmission during the period of communicability, which can be up to 6 months for males.
 - If planning pregnancy, women with ZIKV should wait at least 8 weeks after symptom onset, and men testing positive should wait at least 6 months after symptom onset
 - ZIKV cases with pregnant partners should use condoms for the duration of the pregnancy or 6 months, whichever is longer.
 - Pregnant women with ZIKV should be referred to a maternal fetal medicine specialist and/or an infectious diseases specialist.⁽¹⁰⁾
 - ZIKV cases should defer donating blood for 56 days after the onset of symptoms.⁽²⁵⁾

Management of Contacts

- Generally, contact follow-up is not required as the risk of person-to-person transmission is low.
- If made aware of any pregnant sexual contacts, ensure they are educated about:
 - the importance of using condoms during sex for the duration of pregnancy to prevent the transmission of infection,
 - the signs and symptoms of ZIKV, and
 - consulting their healthcare provider if symptoms do occur.
- Infants born to ZIKV-infected mothers may under-go further evaluation as per their health care provider.

Preventive Measures

- There are currently no vaccines to prevent the acquisition of ZIKV. Research is underway.

Travellers going to ZIKV-Affected Areas

- Provide the following recommendations:
 - Consult a health care professional or visit a travel health clinic preferably six weeks before you travel.
 - Pregnant women and those planning pregnancy should consider avoiding, or postponing, travel to ZIKV-affected areas.⁽²⁶⁾
 - Protect yourself from mosquito bites:⁽¹²⁾
 - remain in well-screened or completely enclosed, air-conditioned areas whenever possible,
 - wear light-colored clothing and cover up as much as possible with long pants, long-sleeved loose fitting shirts, shoes/boots (not sandals) and a hat.
 - sleep under a bed net, preferably treated with insecticide, and
 - use insect repellent (containing DEET or Icaridin), 15 minutes after sunscreen, on exposed skin and on clothing.
 - Protect yourself from sexually transmitted ZIKV by using a condom when having sex.

All Travellers returning from ZIKV-Affected Areas^(10,26)

- Men should use condoms during sex, wait to conceive and defer all semen donations for 6 months after their return, even if no symptoms occur.
- If the returning traveller has a pregnant partner, condoms (male or female) should be used during sex for the duration of the pregnancy. Further studies are required to determine if other barrier methods (e.g., cervical cap, diaphragm) are adequate protection against ZIKV transmission.⁽⁹⁾
- Women should use condoms during sex and delay conception for 2 months after their return, even if no symptoms occur.
- Blood and tissue donations should be deferred for a period of 21 days following travel to ZIKV outbreak countries.⁽²⁾

Pregnant Travellers returning from ZIKV-Affected Areas⁽²⁷⁾

- Pregnant women should seek medical attention if the signs and symptoms compatible with acute ZIKV occur while travelling or upon return home.
- Pregnant women may under-go further evaluation as per their health-care provider.
- The decision to test for ZIKV without the presence of symptoms should be discussed between the pregnant woman and her health care provider.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
April 2018	Guideline	Previous guideline update including case definition, reporting requirements, epidemiology and public health management
September 2021	Case Definition	- Addition of IgG seroconversion to confirmed case definition - Footnote added regarding change to dengue PRNT testing
	Lab interpretation Table	Updated to reflect change to case definition
	Reporting Requirements	Removed "in the prescribed form" from physicians, health practitioners and others section.
	Diagnosis	Moved to Epidemiology section and updated to reflect changes in testing
	Treatment	Moved to Epidemiology section
August 2022	General	- Template updated - Web links refreshed
	Table 1	Added equivocal result to Dengue testing as per new testing protocol at ProvLab.

References

1. Rabe IB, Staples JE, Villanueva J, Hummel KB, Johnson JA, Rose L, et al. Interim Guidance for Interpretation of Zika Virus Antibody Test Results. *MMWR Morb Mortal Wkly Rep*. 2016;65(21):543–6.
2. Government of Canada. Rapid risk assessment: The risk of Zika virus to Canadians (first update) [Internet]. 2016. Available from: <https://www.canada.ca/en/public-health/services/publications/diseases-conditions/rapid-risk-assessment-risk-zika-virus-canadians-first-update.html>
3. Government of Canada. Zika virus: Symptoms and treatment [Internet]. 2019. Available from: <https://www.canada.ca/en/public-health/services/diseases/zika-virus.html?id=symptoms>
4. Broutet N, Brauer F, Riesen M, Khalakdina A, Almiron M, Aldighieri S, et al. Zika virus as a cause of neurologic disorders. *N Eng J Med*. 2016;374(16):DOI: 10.1056/NEJMp1602708.
5. Cao-Lormeau V-M, Blake A, Mons S, Lastère S, Roche C, Vanhomwegen J, et al. Guillain-Barré Syndrome outbreak associated with Zika virus infection in French Polynesia: a case-control study. *Lancet* [Internet]. 2016;387(10027):1531–9. Available from: linkinghub.elsevier.com/retrieve/pii/S0140673616005626
6. World Health Organization (WHO). Zika virus and complications: Questions and answers. World Health Organization; 2018.
7. Rasmussen SA, Jamieson DJ, Honein MA, Petersen LR. Zika Virus and Birth Defects — Reviewing the Evidence for Causality. *N Engl J Med*. 2016;1–7.
8. ProVLab. Update on testing for Zika, Dengue and Chikungunya Viruses [Internet]. 2016. Available from: www.provlab.ab.ca/partner_updates.htm
9. Raymond T. Personal Communication. Edmonton, AB: Provincial Laboratory for Public Health (ProvLab); 2017.
10. Committee to Advise on Tropical Medicine and Travel. Zika Virus Prevention and Treatment Recommendations [Internet]. Government of Canada. 2020. Available from: https://www.canada.ca/en/public-health/services/publications/diseases-conditions/zika-virus-prevention-treatment-recommendations.html?id=zika_virus_16_phac_catmat#s7a
11. Centers for Disease Control and Prevention. Clinical Evaluation & Disease | Zika Virus [Internet]. 2017. Available from: www.cdc.gov/zika/hc-providers/preparing-for-zika/clinicalevaluationdisease.html
12. Government of Canada. Zika virus: Prevention and risks [Internet]. 2019. Available from: <https://www.canada.ca/en/public-health/services/diseases/zika-virus/prevention-risks.html>
13. World Health Organization (WHO). Zika virus [Internet]. World Health Organization; 2016. Available from: www.who.int/mediacentre/factsheets/zika/en/
14. Centers for Disease Control and Prevention (CDC). Zika Virus: Transmission and Risks [Internet]. 2016. Available from: www.cdc.gov/zika/transmission/
15. Centers for Disease Control and Prevention (CDC). CDC assisting Utah investigation of Zika virus infection apparently not linked to travel [Internet]. 2016. Available from: www.cdc.gov/media/releases/2016/s0718-zika-utah-investigation.html
16. Müller JA, Harms M, Schubert A, Jansen S, Michel D, Mertens T, et al. Inactivation and Environmental Stability of Zika Virus. *Emerg Infect Dis*. 2016 Sep;22(9):1685–7.
17. European Centre for Disease Control and Prevention. Zika virus disease [Internet]. 2016. Available from: <https://www.ecdc.europa.eu/en/zika-virus-disease>
18. Prisant N, Bujan L, Benichou H, Hayot P-H, Pavili L, Lurel S, et al. Zika virus in the female genital tract. *Lancet Infect Dis* [Internet]. Elsevier; 2016 Sep;16(9):1000–1. Available from: linkinghub.elsevier.com/retrieve/pii/S1473309916301931
19. Musso D, Roche C, Nhan T-X, Robin E, Teissier A, Cao-Lormeau V-M. Detection of Zika virus in saliva. *J Clin Virol* [Internet]. 2015 Jul;68:53–5. Available from: <http://www.ncbi.nlm.nih.gov/pubmed/26071336>
20. Gourinat A-C, O'Connor O, Calvez E, Goarant C, Dupont-Rouzeyrol M. Detection of Zika Virus in Urine. *Emerg Infect Dis* [Internet]. 2015 Jan;21(1):84–6. Available from: wwwnc.cdc.gov/eid/article/21/1/14-0894_article.htm

21. Nicastrì E, Castillettì C, Liuzzi G, Iannetta M, Capobianchi MR, Ippolito G. Persistent detection of Zika virus RNA in semen for six months after symptom onset in a traveller returning from Haiti to Italy, February 2016. *Euro Surveill Bull Eur sur les Mal Transm = Eur Commun Dis Bull* [Internet]. 2016;21(32):1–4. Available from: www.ncbi.nlm.nih.gov/pubmed/27541989
22. Alberta Health. Communicable Disease Reporting System (CDRS). Edmonton, AB; 2017.
23. Heymann DL, editor. *Control of Communicable Diseases Manual*. 20th ed. Washington, DC: American Public Health Association; 2015.
24. Centers for Disease Control and Prevention (CDC). Zika Virus Disease and Zika Virus Infection | 2016 Case Definition [Internet]. 2016. Available from: wwwn.cdc.gov/nndss/conditions/zika/case-definition/2016/06/
25. Canadian Blood Services. Personal Communication. 2016.
26. Government of Canada. Travel health notices – Zika virus: Advice for travellers [Internet]. 2019. Available from: https://travel.gc.ca/travelling/health-safety/travel-health-notices/152?_ga=1.112940098
27. Government of Canada. Zika virus: Information for health professionals [Internet]. 2018. Available from: <https://www.canada.ca/en/public-health/services/publications/diseases-conditions/zika-virus-information-health-professionals.html>